Customer Wait-Time Improvement Project

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Project Overview

Problem: During the lunch rush, the drive-through often backs up. Customers with quick transactions then come inside, where they wait in the same line as customers with long transactions.

Goal: Reduce lobby wait times **without adding or moving employees**.

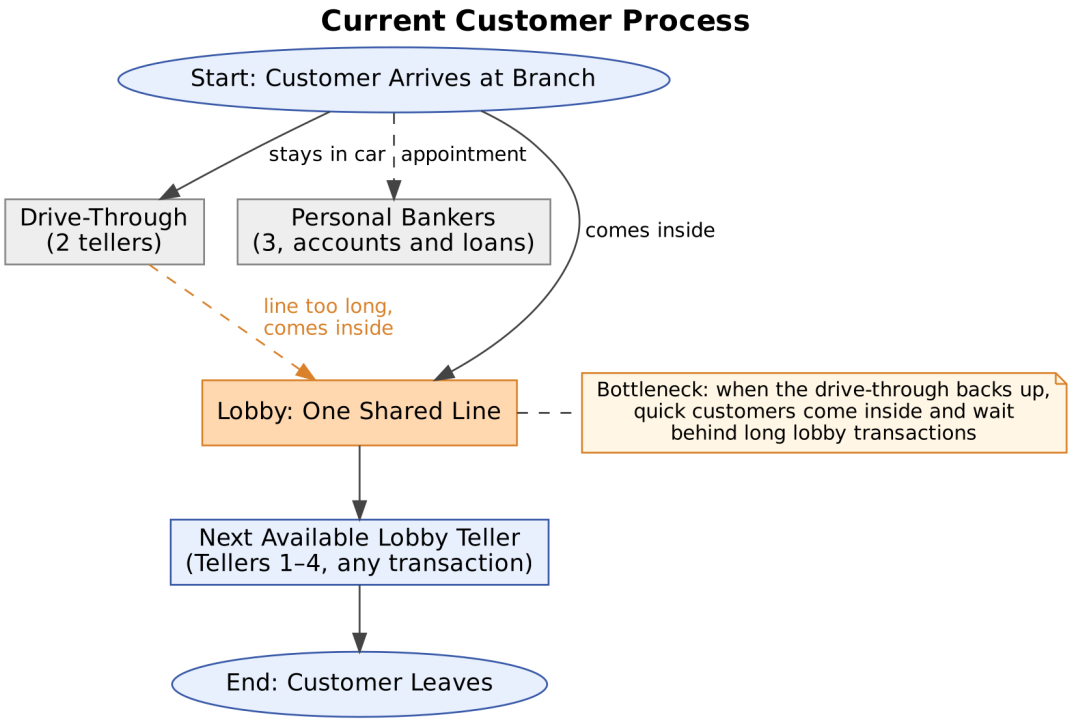
Setting: A bank branch staffed with 4 lobby tellers, 2 drive-through tellers, and 3 personal bankers. This project changes only how the 4 lobby tellers are assigned. Drive-through tellers and personal bankers keep their normal duties.

Method: Map the current process, simulate a lunch rush in a spreadsheet, identify the bottleneck, and compare two alternative designs against the current process using the same staff and the same customers.

Tools: Google Docs, Google Sheets, diagrams.net.

Data and confidentiality: All customer data is simulated for educational purposes. No real customer information, transaction records, account data, or internal bank procedures were used, and the bank is not named. Only general staffing levels were used to make the scenario realistic.

Current Process Map



Causes of Waiting

1. **Peak demand and drive-through overflow:** when the drive-through line is long, quick-transaction customers come inside, adding to the lobby rush.

2. **Long service time:** long transactions (12 minutes) occupy lobby tellers while quick customers (3 minutes) wait behind them in the same line.

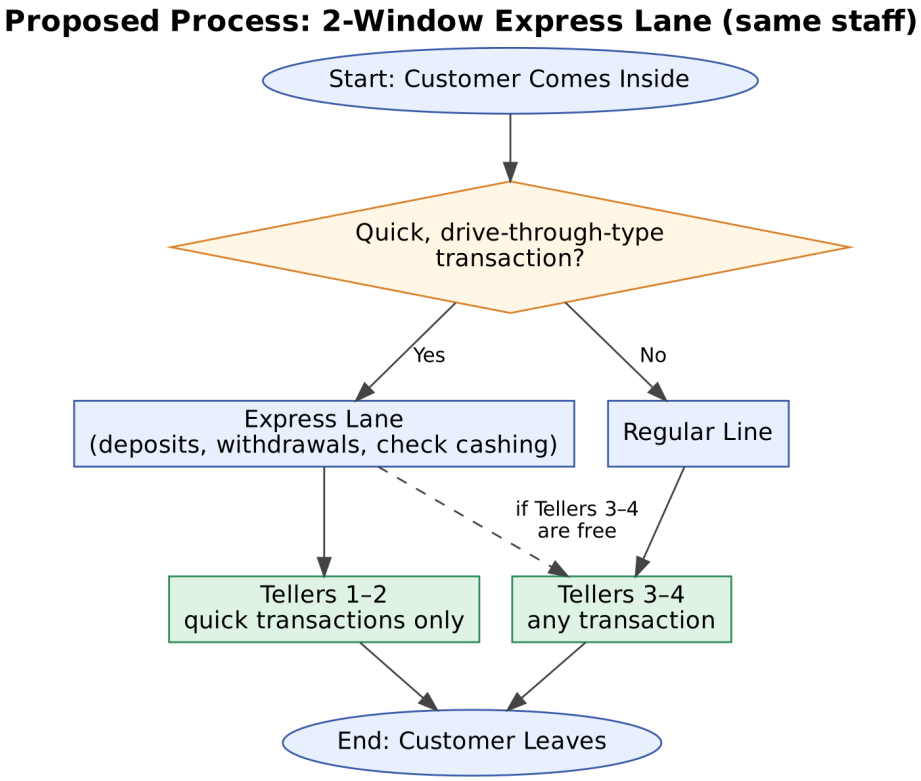
Simulated Scenario

Sixteen customers come inside between 12:00 and 12:08: twelve with quick transactions (deposits, withdrawals, check cashing; 3 minutes each) and four with long transactions (12 minutes each). Each teller serves the first customer in line that they are allowed to serve.

Design Alternatives Compared

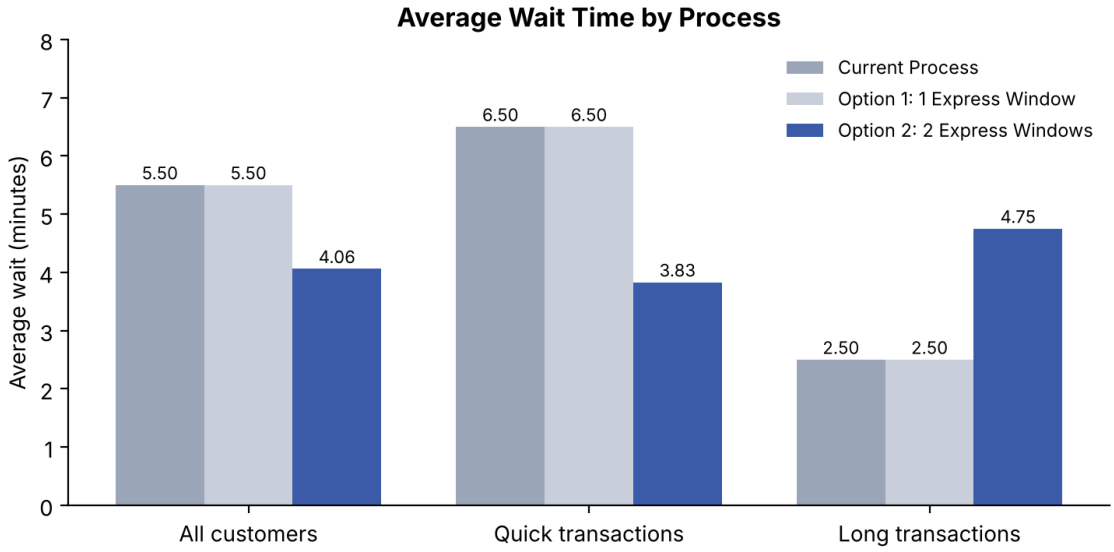
Design	Tellers 1–2	Tellers 3–4
Current Process	Any transaction (one shared line)	Any transaction
Option 1: 1 Express Window	Teller 1: quick only; Teller 2: any	Any transaction
Option 2: 2 Express Windows	Quick transactions only (express lane)	Any transaction (serve express customers when free)

Proposed Process (Option 2)



Results

Measure	Current	Option 1	Option 2
Average wait, all customers	5.50 min	5.50 min	4.06 min (-26%)
Average wait, quick transactions	6.50 min	6.50 min	3.83 min (-41%)
Average wait, long transactions	2.50 min	2.50 min	4.75 min (+2.25)
Longest single wait	11 min	11 min	10 min



Key Findings

One express window did not help. In the shared line, the tellers who were not tied up with long transactions were already serving the quick customers, so reserving one window only changed which teller did that work, not how long anyone waited. This shows why testing alternatives matters: an idea that sounds good may not change the result.

Two express windows made a real difference. Quick-transaction customers, who are most of the lunch rush, waited 41% less, the overall average dropped 26%, and the longest wait also fell slightly.

Trade-off: Customers with long transactions waited about 2 minutes longer on average, because only 2 tellers serve them. Since these customers are often already planning for a longer visit, this trade-off may be acceptable, but it should be monitored.

Conclusion and Recommendation

The main cause of lobby waits during the lunch rush is quick-transaction customers, many of them drive-through overflow, waiting behind long transactions. A 2-window express lane reduced average wait by 26% overall and 41% for quick transactions using the same 4 lobby tellers, with no change to drive-through or personal banker staffing. Recommended next steps: (1) let express tellers serve the regular line when the express lane is empty, (2) test the design with a larger simulation using random arrival times, and (3) collect anonymous, aggregate timing data (with management approval) to confirm the results.